WANTED

ILL-POSED AUTOREGRESSIVE PROBLEMS

AUTOREGRESSIVE MODELS

AR(P) with a Gaussian prior on the coefficients $\mathbf{a}$ and innovations $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$:

$$x_t = a_1 x_{t-1} + \ldots + a_P x_{t-P} + \varepsilon_t$$
$$\mathbf{a} \sim \mathcal{N}(\mathbf{0}, \lambda \mathbf{I})$$
$$\mathbf{x} \mid \mathbf{a} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma_a})$$

This models a digital LTI system with transfer function

$$H(z) = \frac{1}{A(z)} = \frac{1}{1 - a_1 z^{-1} - \ldots - a_P z^{-P}}$$

with power spectrum $\sigma^2 \; |H(f)|^2$.

THE IDEA

AR(4), impose one resonance $Q(z) \mid A(z)$:

$$A(z) = 1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3} - a_4 z^{-4}$$
$$Q(z) = 1 - cz^{-1} + dz^{-2}, \quad c = 2\rho \cos \omega_0, d = \rho^2$$

Divisibility means a quotient $R(z)$ exists:

$$A(z) = Q(z)\, R(z), \quad R(z) = 1 - r_1 z^{-1} - r_2 z^{-2}$$

Match coefficients, eliminate $\mathbf{r}$: what remains is linear:

$$\mathbf{F}\mathbf{a} + \mathbf{c} = \mathbf{0}, \quad \mathbf{F} = \begin{pmatrix} 1 & 0 & -1/d & -c/d^2 \\ 0 & 1 & c/d & c^2/d^2 - 1/d \end{pmatrix}$$

Impose it *on average*, $\mathbb{E}_{\mathbf{a}}[\mathbf{F}\mathbf{a} + \mathbf{c}] = \mathbf{0}$:

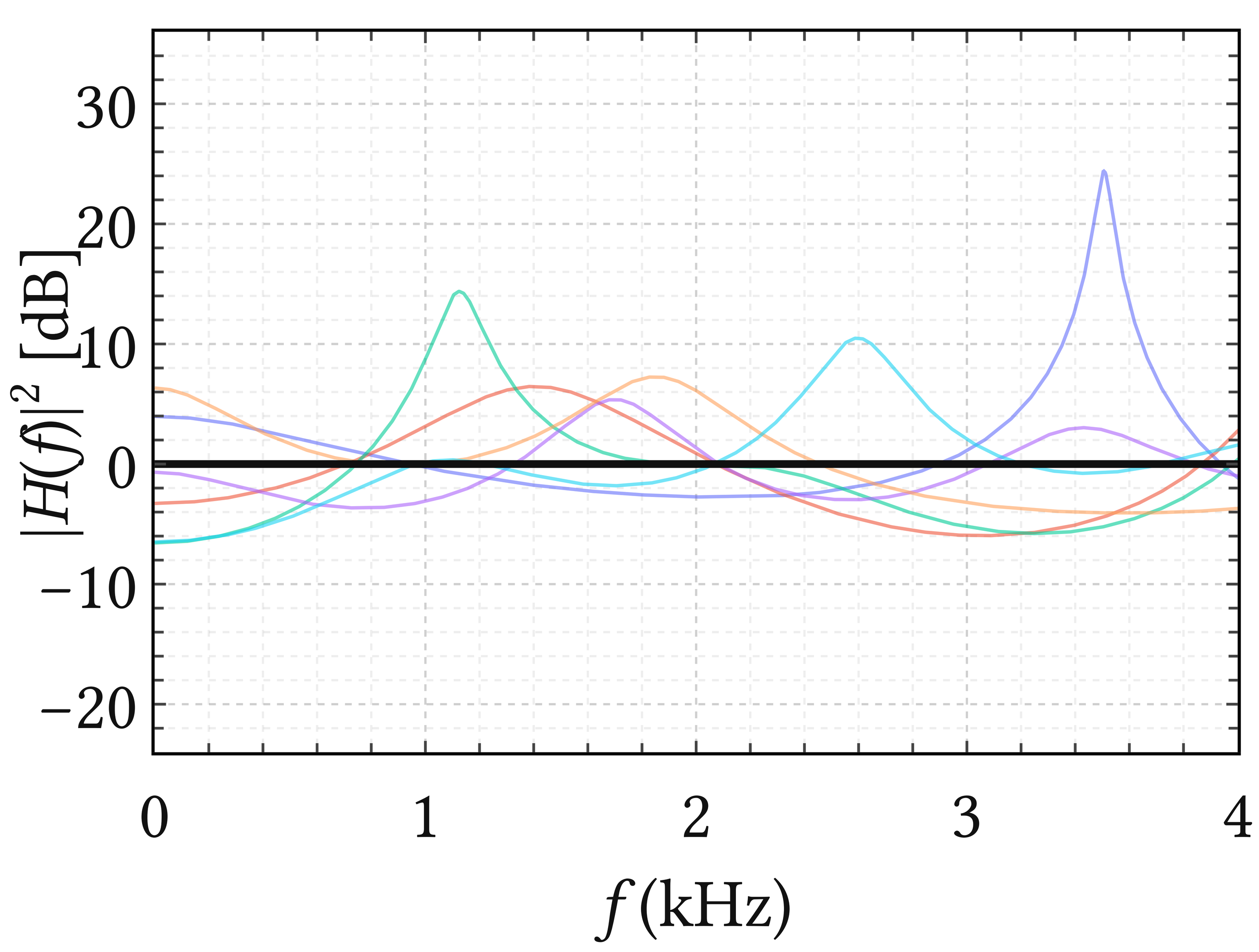
$$\boldsymbol{\mu}_M = -\mathbf{F}^\top (\mathbf{F}\mathbf{F}^\top)^{-1} \mathbf{c}$$

AVAILABLE FEATURES

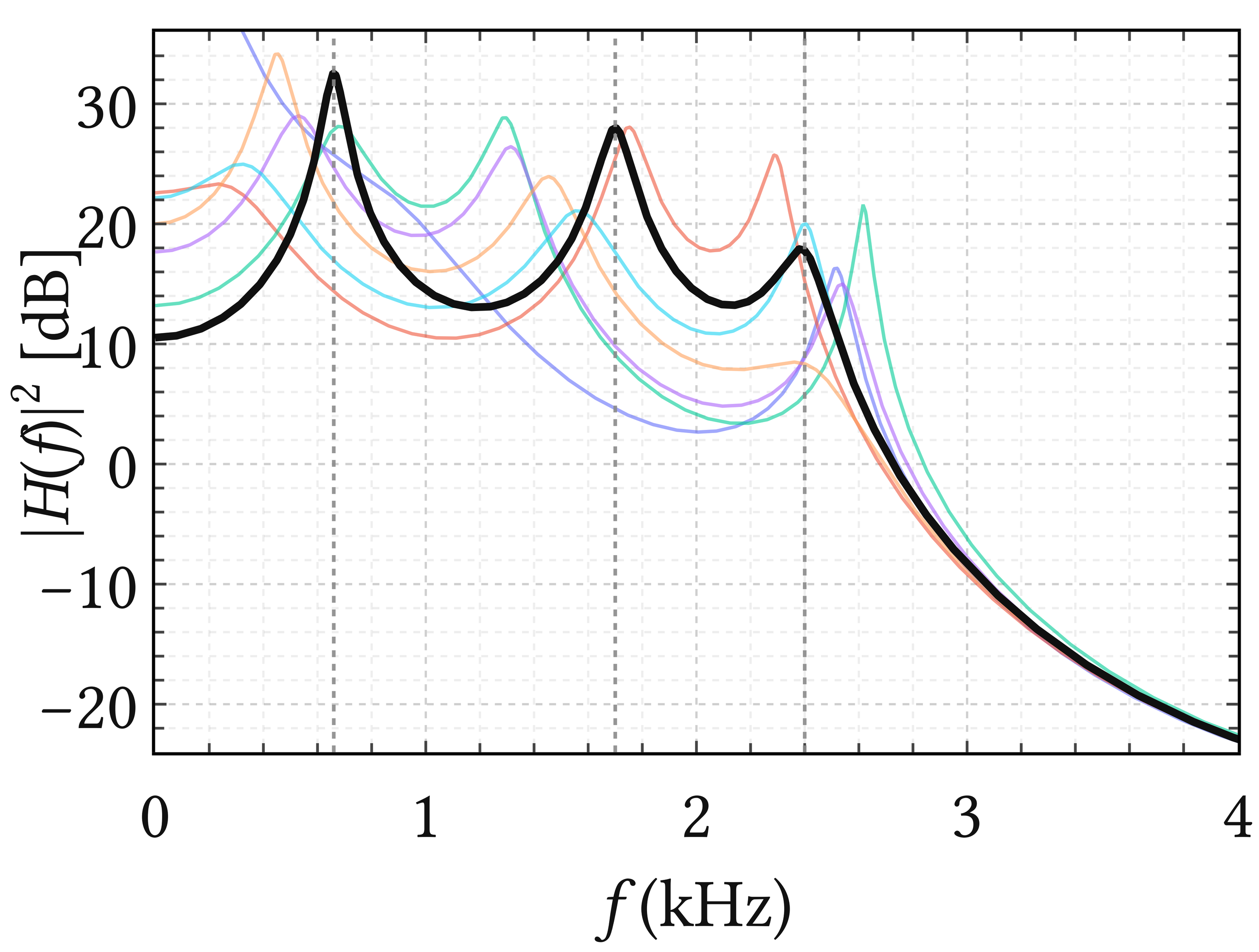
One factor $Q(z)$ per expected feature:

Spectral feature	$Q(z)$	deg
Resonance at (ρ, ω_0)	$1 - 2\rho\cos(\omega_0) z^{-1} + \rho^2 z^{-2}$	2
Real pole at ρ	$1 - \rho z^{-1}$	1
Tilt $-6k$ dB/oct	$(1 - \rho z^{-1})^k$	k
Seasonal pole of period s	$1 - z^{-s}$	s
k-fold resonance	$(1 - 2\rho\cos(\omega_0) z^{-1} + \rho^2 z^{-2})^k$	$2k$

standard prior $\quad \mathbf{a} \sim \mathcal{N}(\mathbf{0}, \lambda \mathbf{I})$



spectral prior $\quad \mathbf{a} \sim \mathcal{N}(\boldsymbol{\mu}_M, \lambda \mathbf{I})$



REWARD Strongly regularized autoregressive problems

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interactive figures